

SAMADHAN SANGARAM BODHANE

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PROFESSIONAL SUMMARY

Electronics and Computer Engineering student at MIT Chh. Sambhajinagar (CGPA 8.92/10) with hands-on experience building full-stack web applications using React, Node.js, Express.js, and Firebase, alongside embedded systems projects involving Arduino, sensor integration, and hardware communication protocols. Delivered a production-deployed AI-integrated productivity platform and a hardware fire-detection system. Experienced in OpenAI API integration, real-time databases, and REST API design. Seeking software engineering, full-stack, or embedded systems roles.

TECHNICAL SKILLS

Programming Languages	C, Python, Java, MATLAB, Verilog / VHDL
Web Technologies	HTML5, CSS3, JavaScript (ES6+), REST APIs
Frameworks & Libraries	React.js, Node.js, Express.js
Databases	MySQL, MongoDB, Firebase Realtime Database
Embedded & EDA Tools	Arduino IDE, Xilinx Vivado, Proteus, Keil uVision
Protocols	UART, Ethernet/TCP-IP, UDP, SNMP
Developer Tools	VS Code, Git, GitHub, Vercel
Cloud & Deployment	Firebase, Vercel
APIs & Integrations	OpenAI API
Core CS Concepts	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROJECTS

ApexLink — Full-Stack Productivity Web Application

React.js · Node.js · Express.js · Firebase · OpenAI API · Vercel | 2024-2025

- Architected a full-stack productivity platform with a unified dashboard for task management, habit tracking, AI-powered timetable generation, and team collaboration using React.js frontend and Node.js/Express.js backend.
- Integrated Firebase Realtime Database and Authentication to enable secure login, real-time data sync across sessions, and role-based access to collaborative workspaces.
- Built VedAI, an embedded AI assistant powered by the OpenAI API, delivering personalised productivity insights, graph interpretation, and goal-driven timetable recommendations.
- Implemented analytics module with dynamic charts visualising task completion rates, habit streaks, and performance trends to enable data-driven user decision-making.
- Deployed on Vercel with continuous integration pipeline ensuring fast global delivery and zero-downtime production updates.

Arduino-Based Fire Detection and Alert System

Arduino Uno · Flame Sensor · MQ-2 Smoke Sensor · Embedded C · Proteus | 2024

- Designed a real-time fire detection system using Arduino Uno with flame and MQ-2 smoke sensors, detecting hazards with low latency through threshold-based sensor fusion logic.
- Programmed embedded C for multi-sensor data acquisition and state-machine alarm logic, triggering buzzer and LED indicators within milliseconds of hazard detection.
- Simulated and validated the complete circuit in Proteus before physical implementation, reducing hardware debugging time and verifying correctness of alarm states.
- Engineered a modular, low-cost hardware architecture deployable in homes, offices, and laboratories with minimal setup requirements.

EDUCATION

B.Tech — Electronics and Computer Engineering

Maharashtra Institute of Technology, Chh. Sambhajinagar

CGPA: 8.92 / 10.0

2023 – 2027 (Expected)

Higher Secondary Certificate (12th)

New High School Dawarkadi, Patthan, Chh. Sambhajinagar

88.40%

2021

Secondary School Certificate (10th)

Pratishthan Mahavidyalaya, Patthan, Chh. Sambhajinagar

80.83%

2023

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

- Competed in National Level **Hack For Impact Hackathon** organised by CSI and ACTS — built and pitched a working solution under a time-constrained, team-based environment.
- Competed in National Level **Build For Bharat Hackathon** organised by CSI, IETE, and IGNITE — developed technology-driven solutions targeting social impact challenges.
- Attended **Hackathon Hacked Workshop** by CSI and ACTS — hands-on training in rapid prototyping, MVP development, and technical pitching.
- Volunteered at **National Level Exhibition DiPEX 2K26** by ABVP and Srijan Trust — coordinated logistics and assisted in managing a large-scale technical exhibition.

LANGUAGES KNOWN

English (Professional Working Proficiency) | Hindi (Fluent) | Marathi (Native)